HOSPITAL MANAGEMENT SYSTEM

ASSIGNMENT # 02

Submission Date (June 10, 2024)

BY GROUP # 9

M. HASEEB HAFEEZ	**23011598-013**
M. HASSAN ARIF	**23011598-051**

SE-108 (Software Engineering)

Semester Spring-2024

BS-II (A) Software Engineering

Submitted to

Ms. Hafsa Shareef Dar

UNIVERSITY OF GUJRAT

A WORLD CLASS UNIVERSITY

HOSPITAL MANAGEMENT SYSTEM

TABLE OF CONTENTS

CONTENTS **PAGE**

1. Introduction..... 03

1.1 Existing systems..... 04

1.2 Comparative Analysis..... 04

2. System Modules..... 05

2.1 Modules..... 05

2.2 stakeholders and Actors..... 06

2.3 Roles and Responsibilities..... 06

2.4 System Workflow Diagram..... 07

3. SRS..... 08

3.1 Questionnaire/ Data collection..... 08

3.2 Functional Requirements..... 10

3.3 Use Case Diagram..... 11

3.4 Use Case Description..... 12

4. Software Design..... 15

4.1 Class Diagram..... 15

5. Testing Strategy..... 16

5.1 Test Case..... 16

6. Development Screenshots..... 18

1. Introduction

What is Hospital Management System:

A Hospital Management System is a software application particularly designed to optimize and automate the organizational, economic and medical processes in a hospital or healthcare facility. It serves as central network that links many modules like registration, bed booking, payment administration, medical history. The system intends to enhance work flow optimization and boosts patient wellbeing and security.

Purpose:

This software will provide the organization more proficient registration for their users and handle appointments and patient medical history. It allows the staff and doctors to customize and view appointment timetable. The objective of this system to automate all information including patient data and hospital particulars.

Who:

Name	Roll No.	Role
M. Haseeb Hafeez	23011598-013	Programmer, administrator, tester, requirement engineer
M. Hassan Arif	23011598-051	Programmer, tester, designer, requirement engineer

Problems:

This new system will solve the following problems seen in old system:

1. **Paper-based Records:**

The old system used a lot of paper for patient records, appointments, and prescriptions. This caused problems like mistakes, lost data, and it was hard to find information.

2. **Security and Privacy Problems:**

Paper records were not safe from being lost, damaged, or seen by unauthorized people.

3. **Scalability and Maintenance Problems:**

The old system struggled to handle more patients and data as it grew.

4. **Data Entry Mistakes:**

Manual data entry often led to mistakes like spelling errors or putting data in the wrong place.

5. **Doctor Administration:**

Our system makes it easy to add, remove doctor information without filling out lots of forms.

1.1 Existing Systems

The Existing Hospital management systems involves traditional /physical process for storing and managing essential record of patient. This is so time taking and requires huge cupboards to store files. Usually inadequate and does not pursue management principles regulations. Forms are usually misplaced during complete reviewing process to ensure that no important data is lost. Doctors have to think of various medicines and usually the do not remember to suggest best alternative at that time. There is always lag in finding and fetching patient records.

1.2 Comparative Analysis with Existing Systems

System 1:

Hospital Management System

Reference

https://www.scribd.com/document/130060854/Hospital-Management-System-Project

System 2:

Health Stack System

Reference:

https://github.com/Jawwad-Fida/HealthStack-System

Module	Our System	System # 1	System # 2
Registration	✓	✓	✓
Doctor Management	✓	✓	✓
Patient Management	✓	✓	✗
Invoice Management	✗	✓	✗

2. System Modules

There are three modules in our hospital management system which are as follows:

1. **Registration**
2. **Doctor Management**
3. **Patient Management**

2.1 Modules

1. Registration

- **Sign up:**
  - Enter Name
  - Enter e-mail
  - Enter Password
- **Login:**
  - Enter e-mail
  - Enter Password

2 Doctor Management

- **Add Doctor:**
  - Allows Admin to add doctor in system.
- **Delete Doctor:**

- Allows Admin to delete doctor from system.

3. **Patient Management**

➢ **Add Patient:**

- Allows admin to add patient in system.

➢ **Delete Patient:**

- Allows admin to delete patient from system.

➢ **Allot Token:**

- Allot Token number to client.

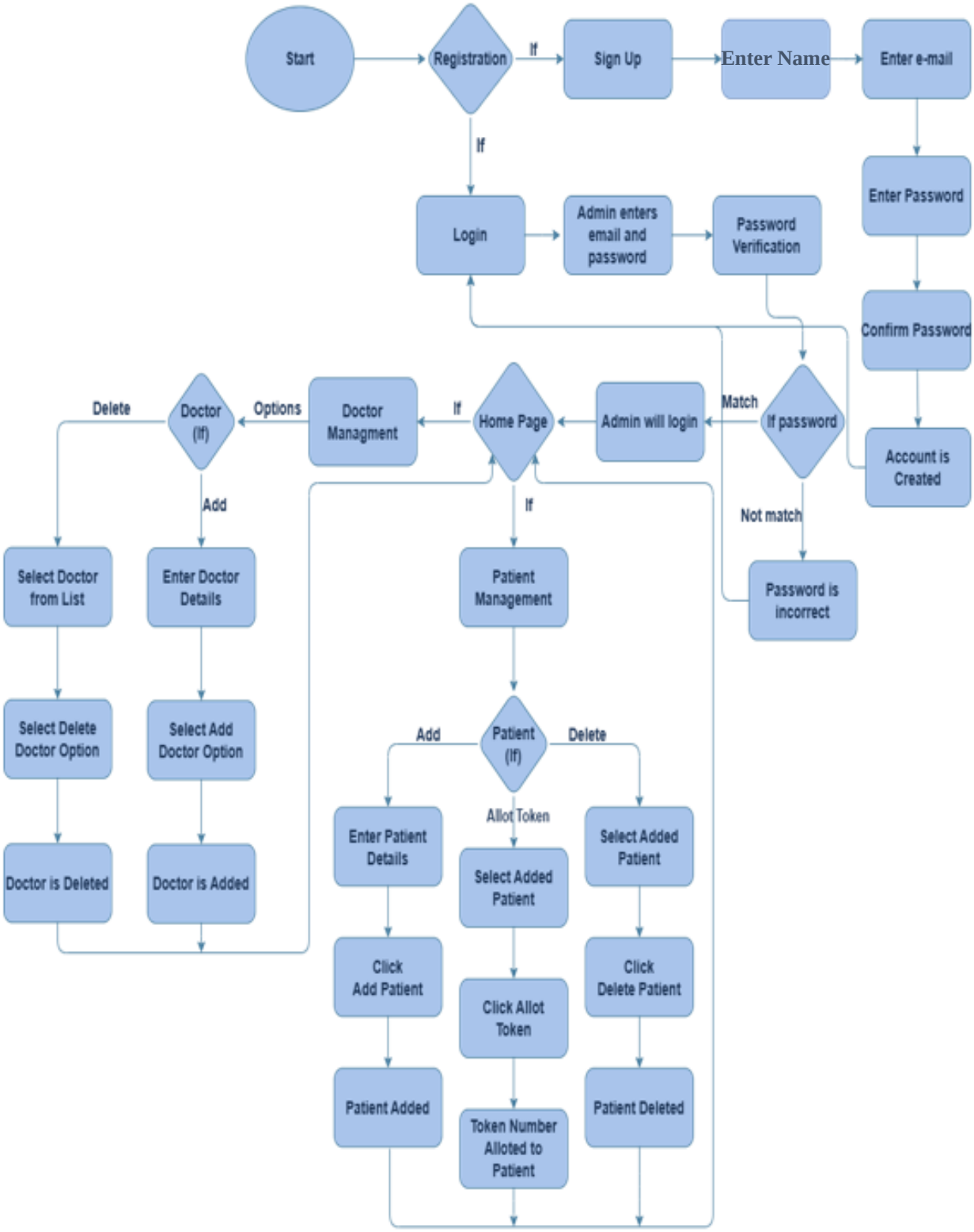
2.2 Stakeholders and Actors

Actors	Stakeholders
Admin **Patient**	**Doctor**

2.3 Roles and Responsibilities

Name	Role	Responsibilities
Admin	**Actor**	These are the persons who play an important role in handling the complete setup. They monitor patient's registration process, manage patient's accounts, add and delete doctor and make sure that the process works efficiently
Patient	**Actor**	These are the people who are associated with appointment booking and can also cancel the appointments as well.

2.3 SYSTEM WORKFLOW DIAGRAM



3. Software Requirement Specification

3.1 Questionnaire/ Data Collection

Question No. 1:

What is the process of adding patient in your hospital management system?

Ans: The system allows the admin to enter patient information such as name, contact information, and disease. After filling the information, the admin adds the patient by clicking on the add patient button.

Question No. 2:

How will the doctor be added in the system?

Ans: In our system, the admin can add a doctor by entering doctor information such as name, specialization and ID. Admins can also delete the previously added doctors.

Question No. 3:

How can the admin create an account?

Ans: The system enables the admin to sign up by entering their name, email address, and password. Admins can then log in using these credentials.

Question No. 4:

How would the system distinguish between doctor/patient and admin roles during the process of signup?

Ans: In our system, only the admin needs to sign up. Doctors and patients are added by the admin into the system, so signup is not required for doctors and patients.

Question No. 5:

How does the admin allot tokens to patients?

Ans: The admin can allot tokens to patients by selecting them from the list. This ensures that the

token allocation is efficient and organized.

Question No. 6:

How does the system manage the display of patient and doctor lists?

Ans: The system uses Jtable to display patient and doctor lists. This provides a tabular format that is easy to read and manage.

Question No. 7:

How does the system handle error messages and notifications?

Ans: The system provides clear error messages and notifications through JOptionPane dialogs. This ensures that users are informed of any issues, such as missing information or unsuccessful operations, and guides them on how to proceed.

Question No. 8:

What steps does system take to validate email and password formats during signup and login?

Ans: If email entered by user during login is not same as email he entered during signup. Then system displays an error message “Invalid email or password”, else user log into system.

Question No. 9:

What process is followed to delete a patient's record from the system?

Ans: The admin selects a patient from the list and press delete button, then the system asks the admin to confirm deletion after confirmation the system delete the patient.

Question No. 10:

Is it possible for admins to delete doctor accounts? Are there any verification steps involved?

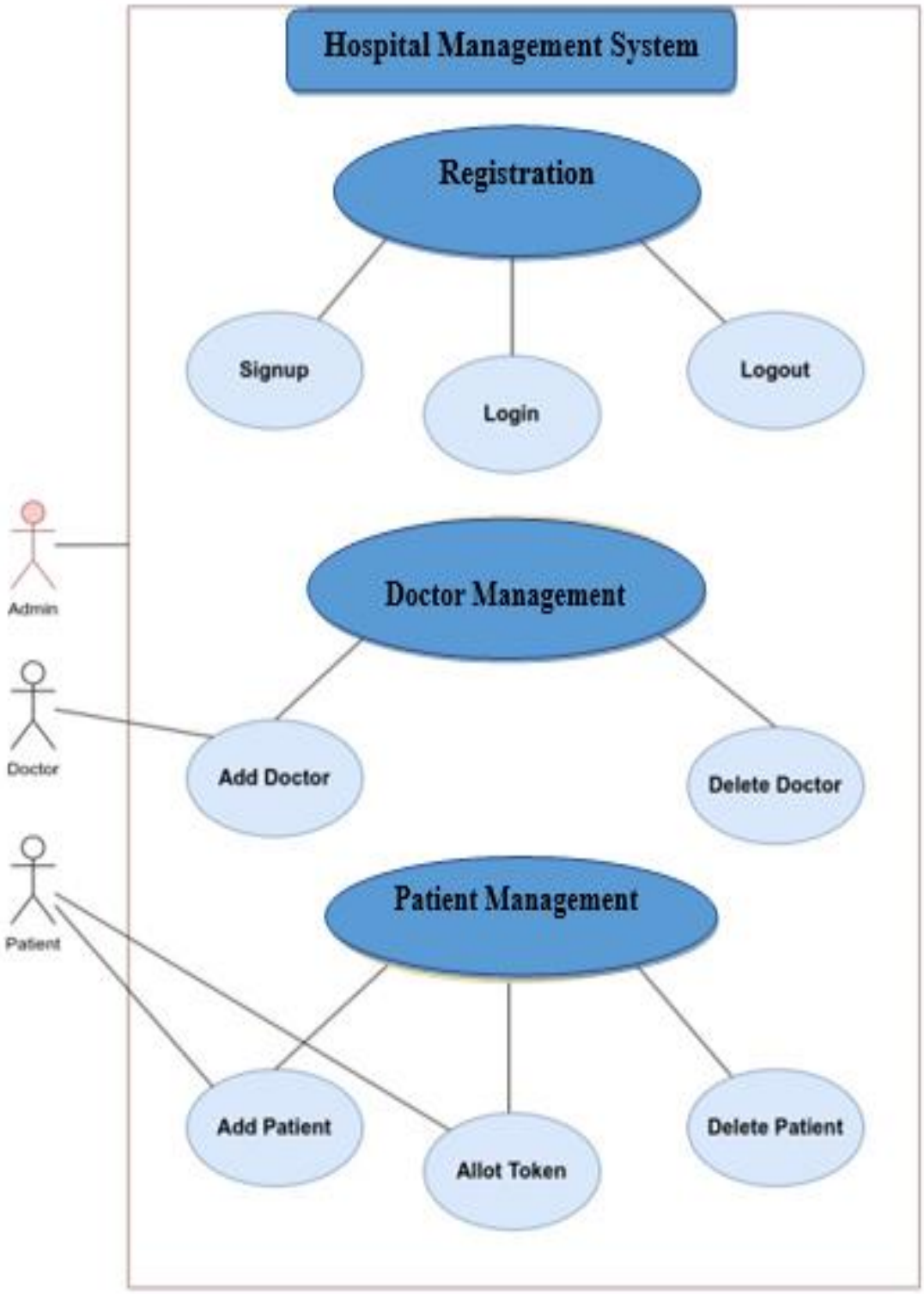
Ans: Yes, the admin can delete doctor accounts. When the admin selects a doctor from the list

for deletion, the system asks for confirmation to "Delete." After confirming, the system shows that the doctor has been successfully deleted.

3.2 Functional Requirements

- **Admin Registration:**
  - Admin can register with the system.
- **Secure Authentication:**
  - Admin are authenticated using a secure login process with strong password policies.
- **Doctor Account Management:**
  - Admin can create, view, edit, and delete doctor accounts.
- **Patient Account Management:**
  - Admin can create, view, edit, and delete patient accounts.
- **Doctor Details Entry:**
  - Admin can enter details like name, specialization, contact information, and qualifications when adding doctors.
- **Patient Data Collection:**
  - Admin can collect essential patient information like name, phone number, and disease during registration.
- **Appointment Management:**
  - Admin can manage appointments between doctors and patients.
  - The system provides a token to identify patients on the day of their appointment.

3.3 Use Case Diagram



3.4 Use Case Description

UC ID: 01	**DEV:** Haseeb, Hassan
UC Name: Delete Doctors	
Actor: Admin	**Date:** June 01, 2024

Pre-Condition:

- Admin should be logged into the system.
- The admin must have navigated to the "Doctor Management" section.
- There should be at least one doctor listed in the system.

Main Structure:

Actors Action:	System Response:
• The admin selects a doctor from the list. displayed in the table by clicking on the desired row.	• The doctor will be selected.
• The admin clicks the "Delete Doctor" button.	• Confirm delete option will appear.
• The admin Confirms the deletion.	• Doctor will be deleted.

Post Conditions:

- Admin has successfully deleted the doctor from system.
- Admin can now go back to Home page.

UC ID: 01 **DEV:** Haseeb, Hassan

UC Name: Allot Token **Date:** June 01, 2024

Actor: Patient

Pre-Condition:

- Admin should be logged into the system.
- The admin must have navigated to the "Patient Management" section.

Main Structure:

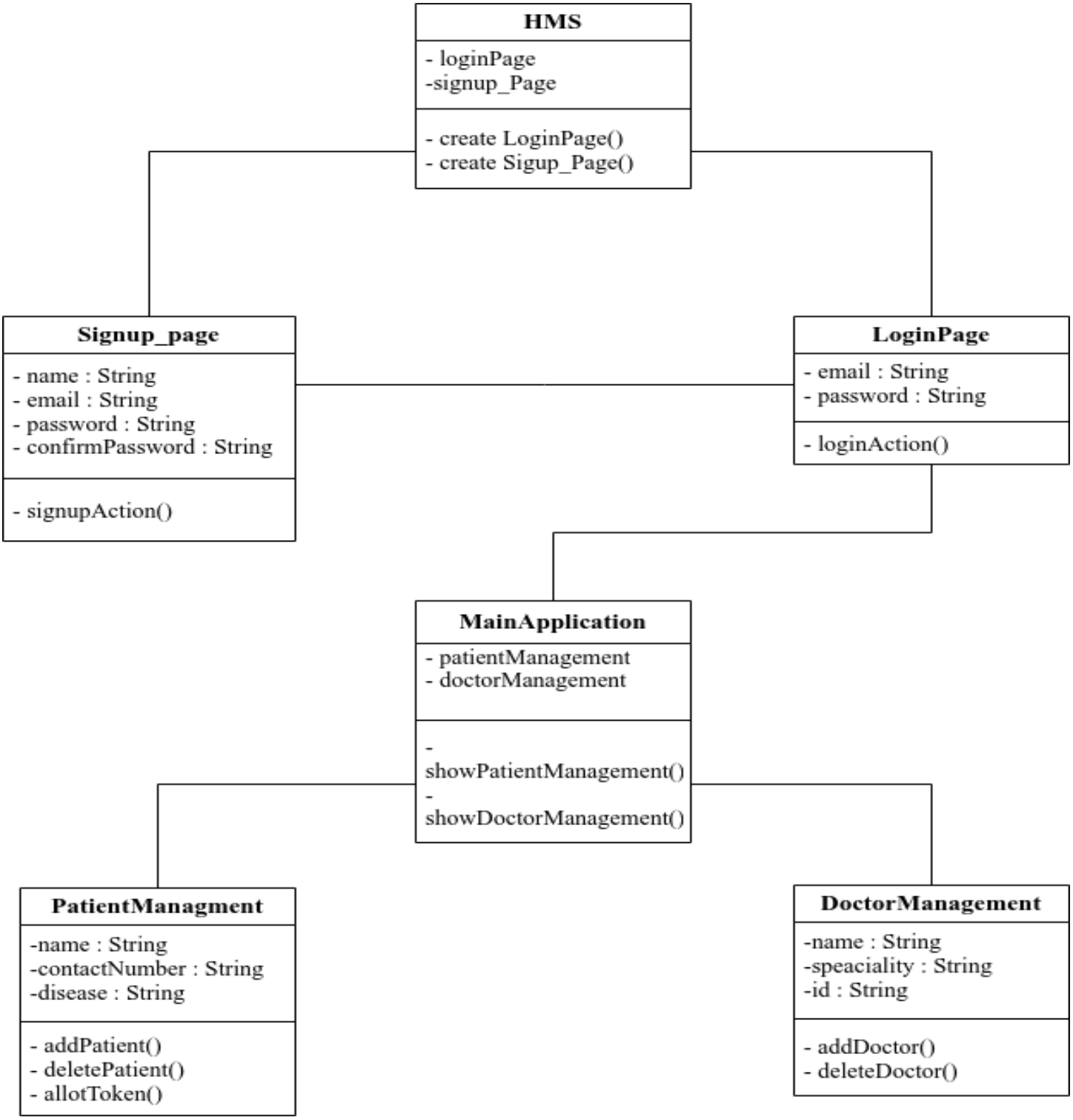
Actors Action:	System Response:
• Admin enter patient details.	
• Admin clicks add patient button.	• The Patient will be added
• Admin select the added patient from list	• Patient will be selected.
• Admin clicks the allot token button	• Token will be allotted to patient.

Post Conditions:

- The selected patient's record shows the assigned token number.
- The token number increases for the next patient.

4. Software Design

4.1 Class Diagram



5. Testing Strategy

5.1 Test Case

T-ID	Function	Test Data	Test Steps	Expected Result	Actual Results	Status
1.1	Add Doctor	**1. Name:** Hassan **2. Specialty:** Skin **3. Doctor ID:** Null	**1.** Put Name **2.** Put Specialty **3.** Put ID **4.** Press Add Doctor Button	System adds the doctor into the system	Error! All fields are required	**Fail**
1.1	Add Doctor	**1. Name:** Hassan **2. Specialty:** Skin **3. Doctor ID:** 4625	**1.** Put Name **2.** Put Specialty **3.** Put ID **4.** Press Add Doctor Button	System adds the doctor into the system	System adds the doctor into the system	**Pass**

T-ID	Function	Test Data	Test Steps	Expected Result	Actual Results	Status
1.2	Allot Token	**1. Name:** Haseeb **2. Contact No:** 0321 **3. Disease:** Eye Infection **4. Add Patient:** Pressed **5. Select Patient:** Null **6. Allot Token:** Pressed	**1.** Put Name **2.** Put Contact No **3.** Put Disease **4.** Press Add patient Button **5.** Select patient from list **6.** Press allots token button	System allots a token to the patient	Error! Please select a patient to allot token	**Fail**
1.2	Allot Token	**1. Name:** Haseeb **2. Contact No:** 0321 **3. Disease:** Eye Infection **4. Add Patient:** Pressed **5. Select Patient:** Selected **6. Allot Token:** Pressed	**1.** Put Name **2.** Put Contact No **3.** Put Disease **4.** Press Add Patient Button **5.** Select patient from list **6.** Press allots token button	System allots a token to the patient	Token is allotted to selected patient	**Pass**

6. Development Screenshots

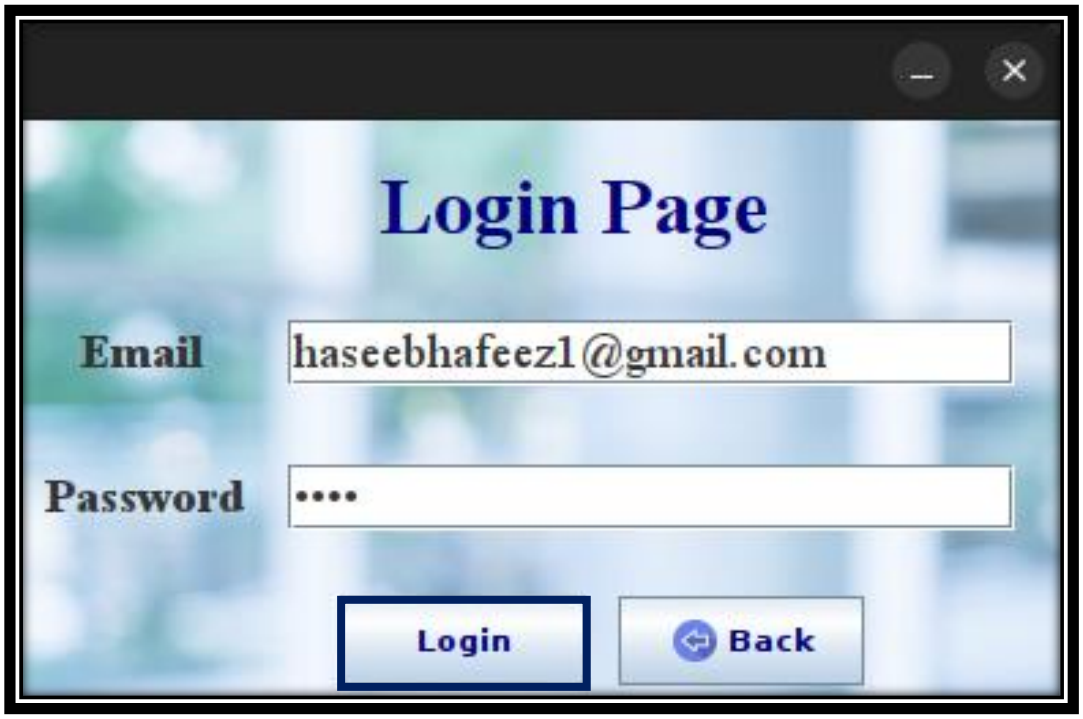
Registration Page

Sign Up Page

Signup Page
Name
Haseeb Hafeez
Email
haseebhafeez1@gmail.com
Password
Confirm Password
Signup
Back

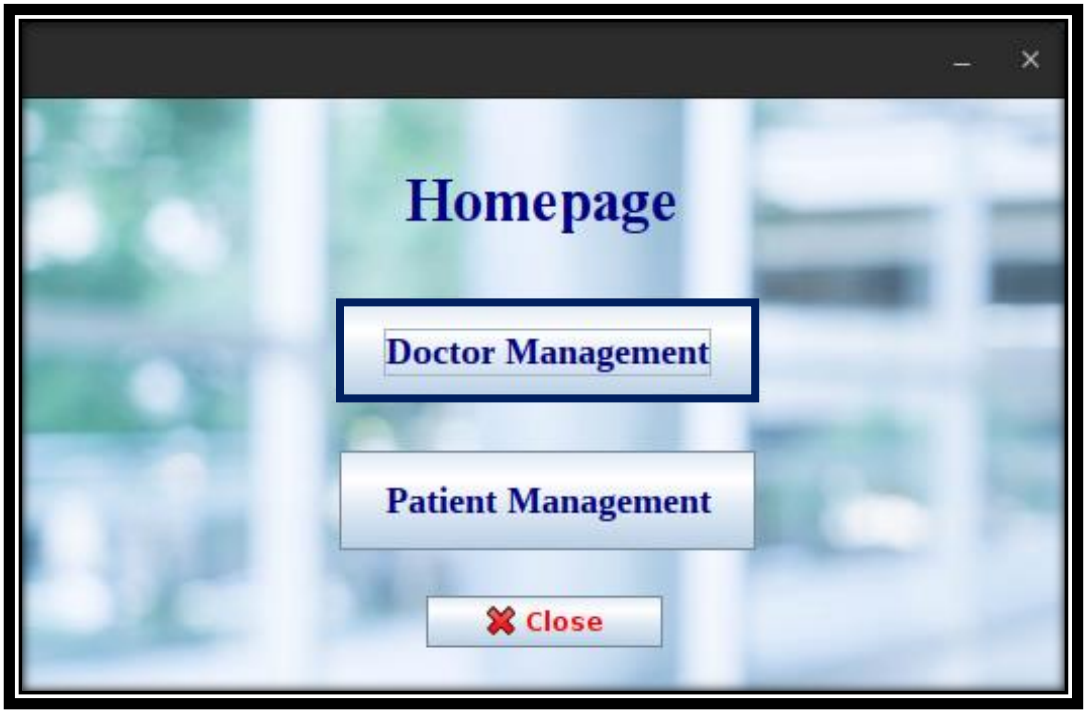
Message
Signup successful! Redirecting to login page...
OK

Login Page

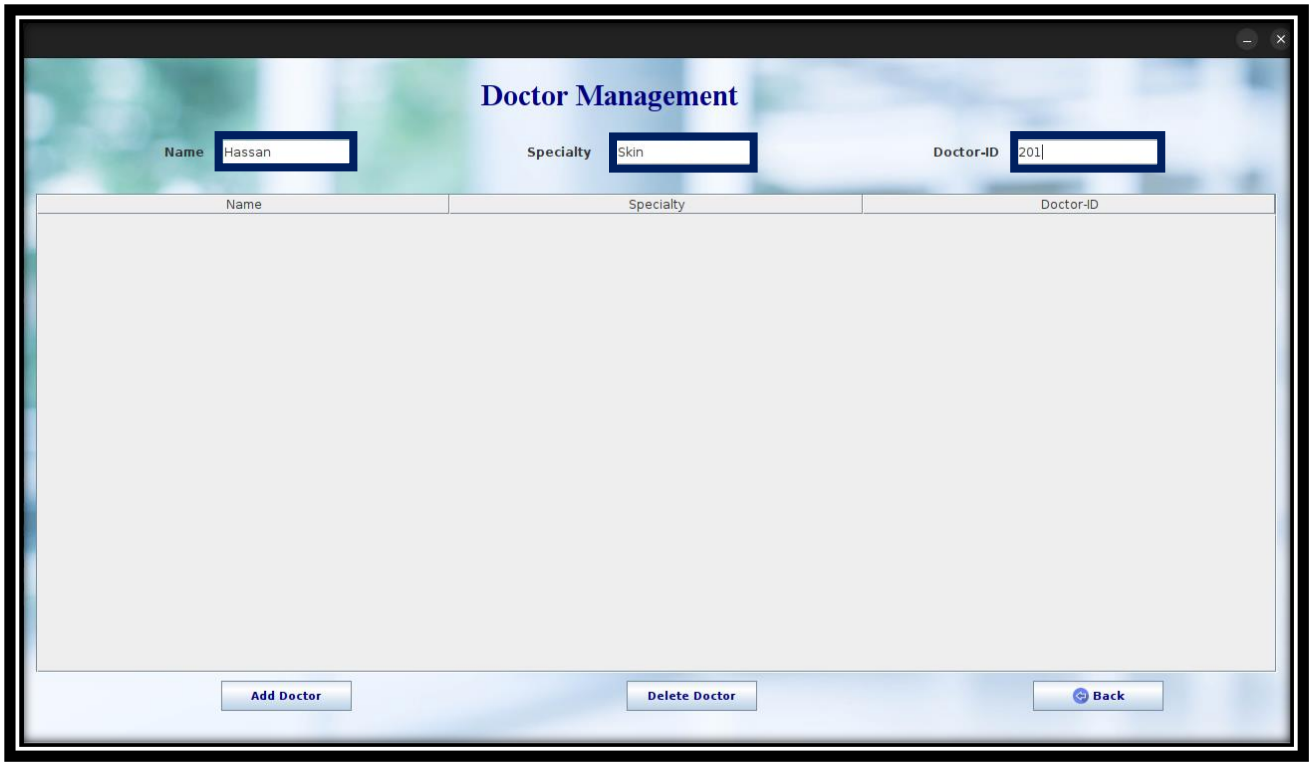

Login Page
Email
haseebhafeez1@gmail.com
Password
Login
Back

Message
Login successful!
OK

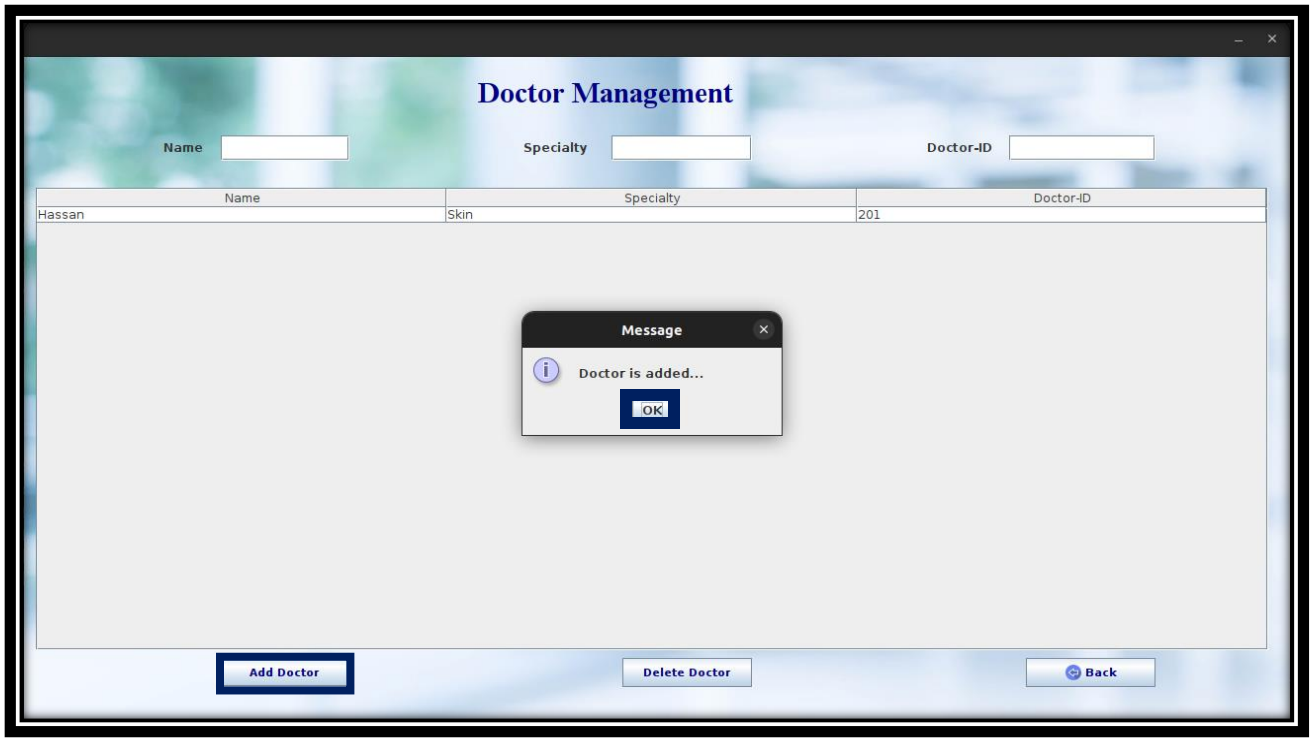
Home Page



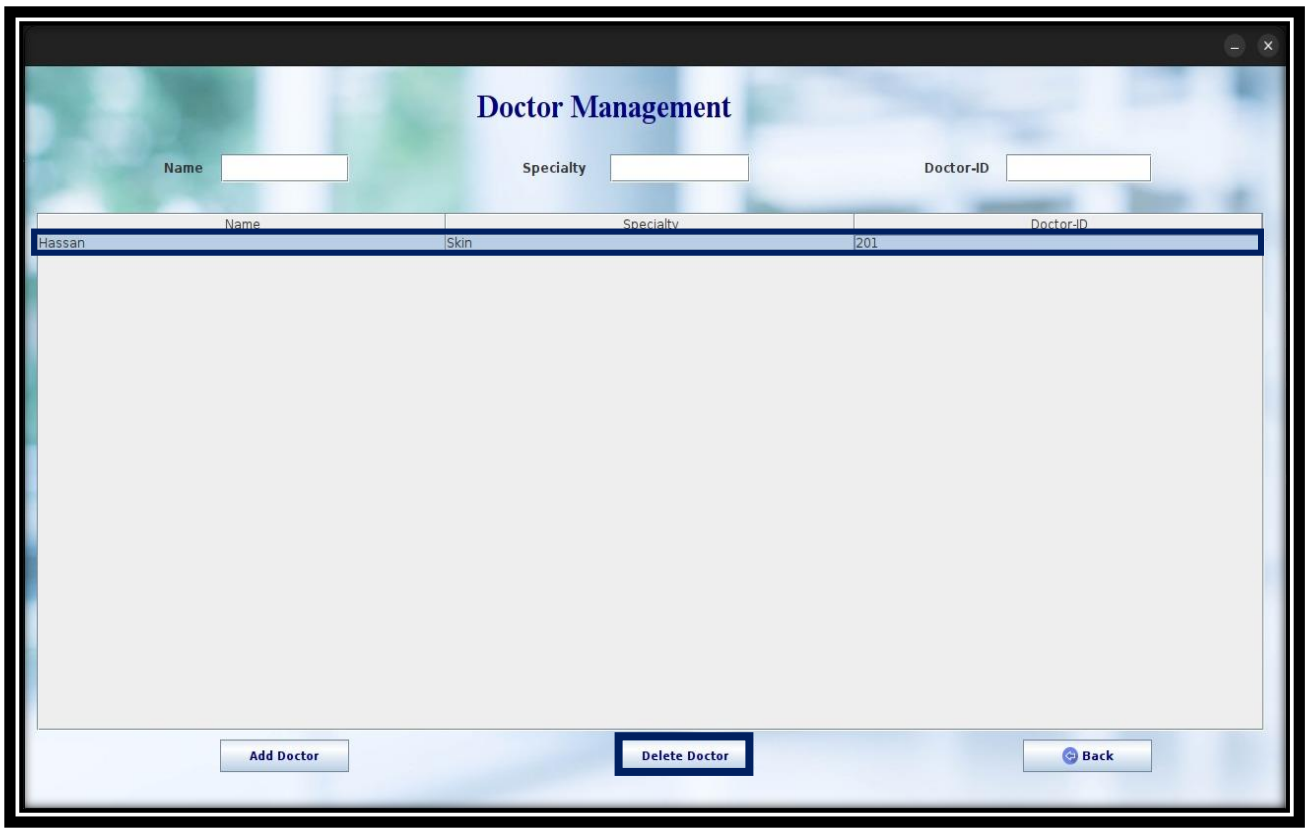
Adding Doctor information



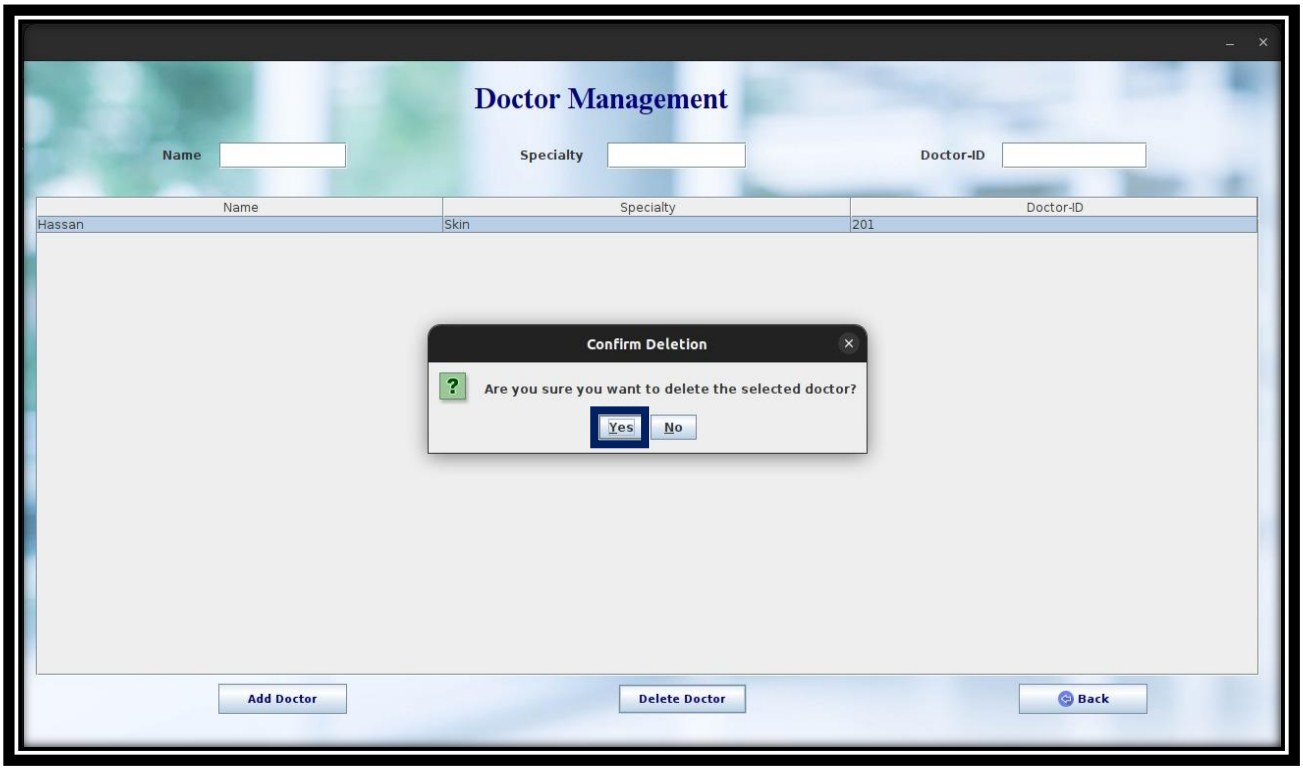
Add Doctor Page



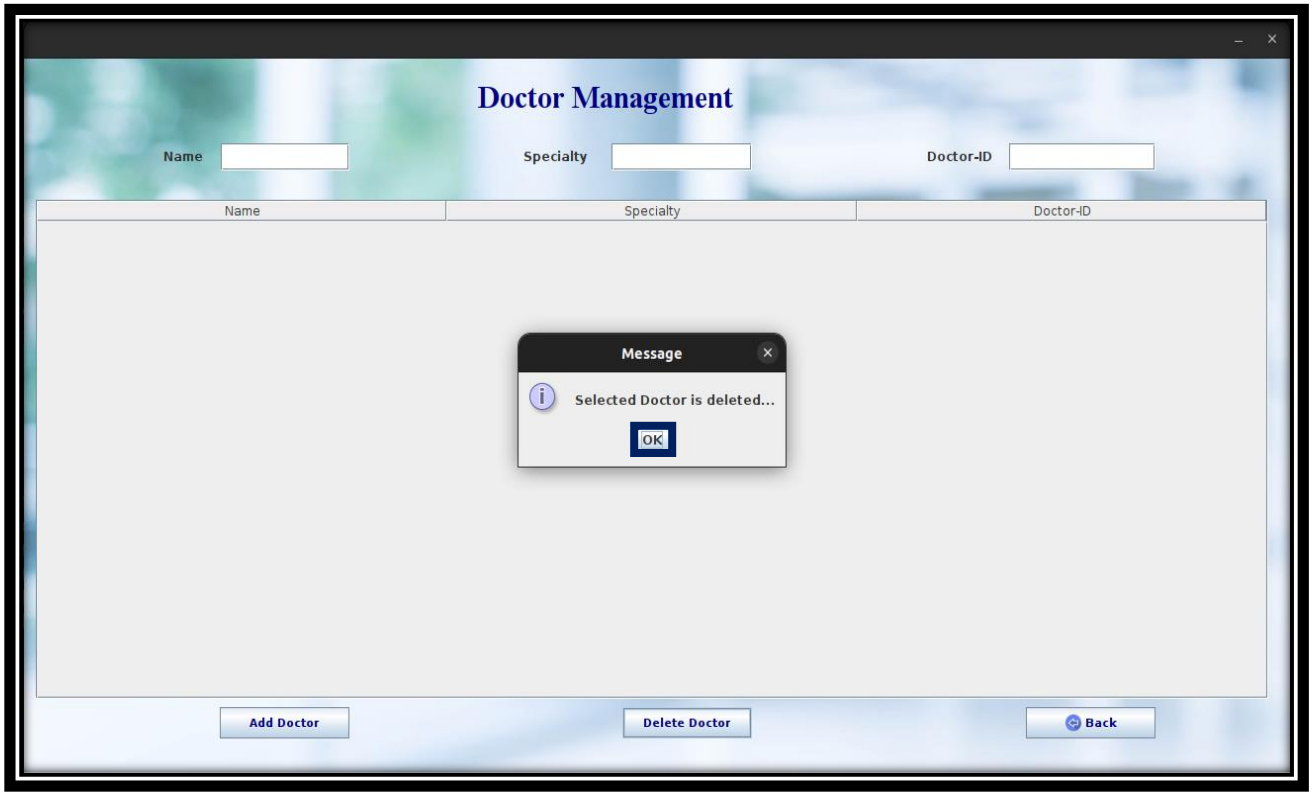
Select Doctor to Delete



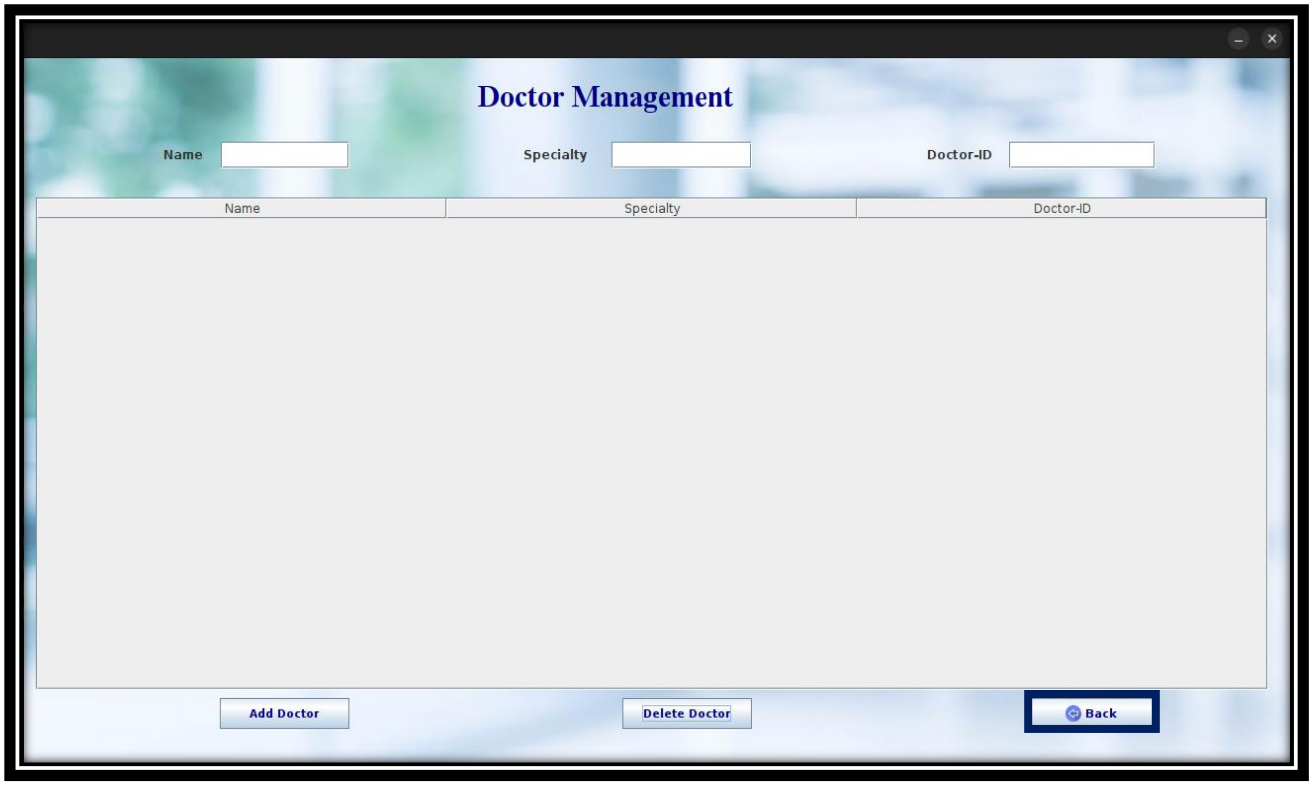
Confirmation Page



Delete Doctor Page

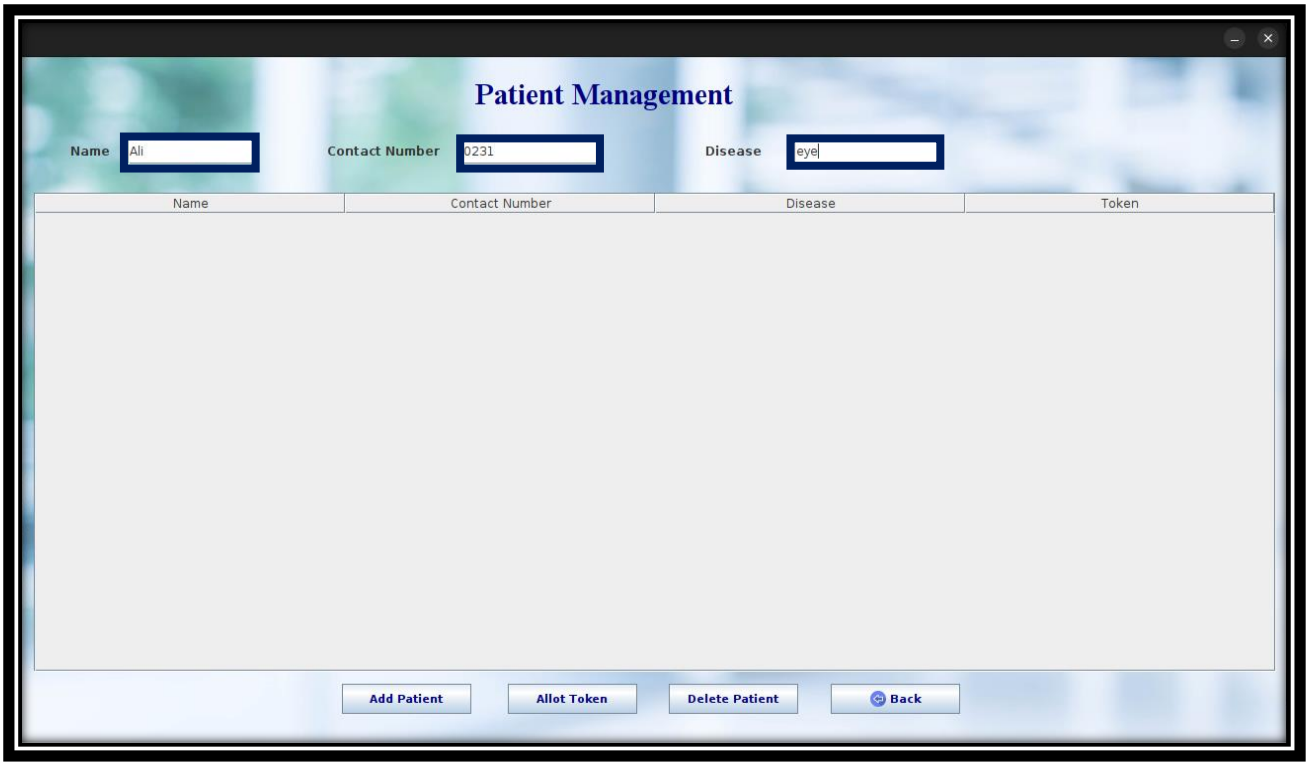


Return to Doctor's Main Page

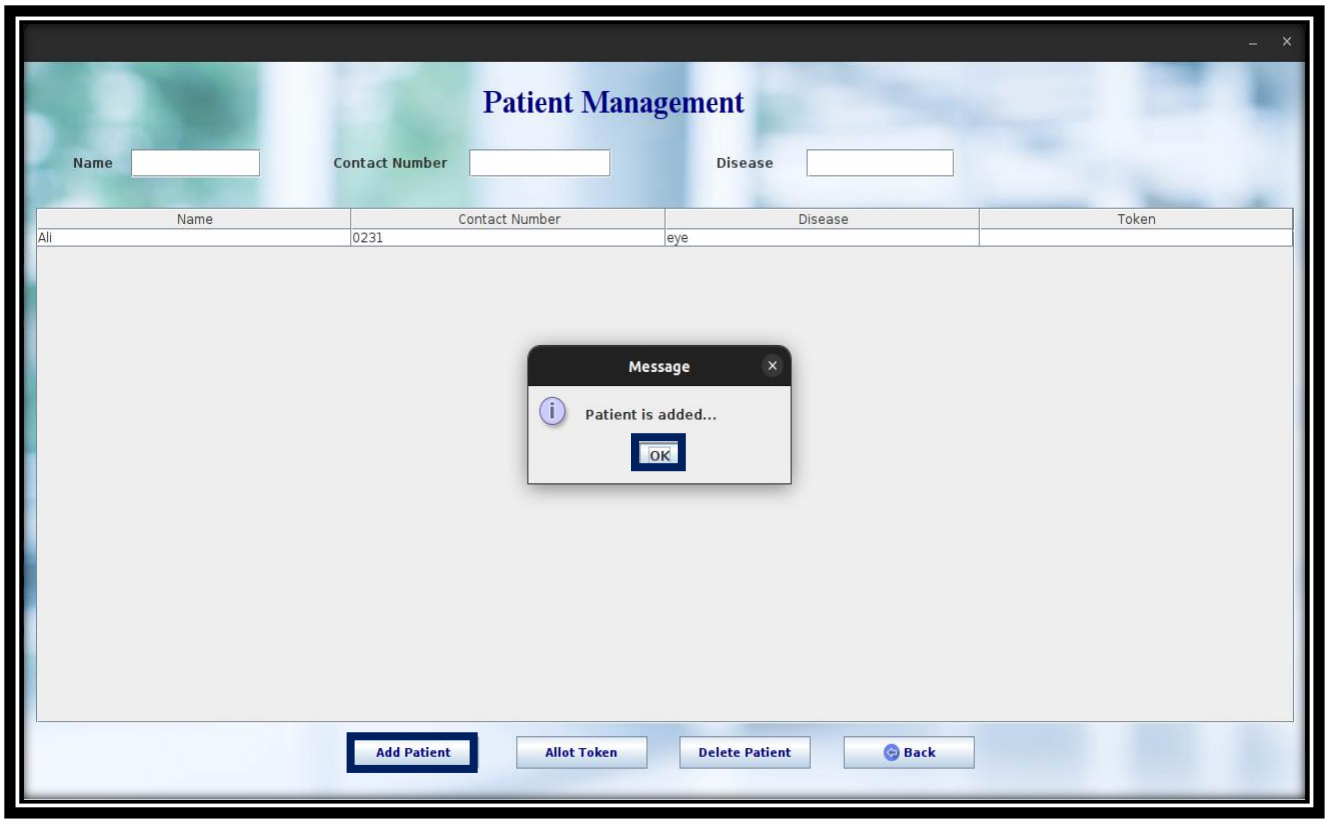


Return to Home Page

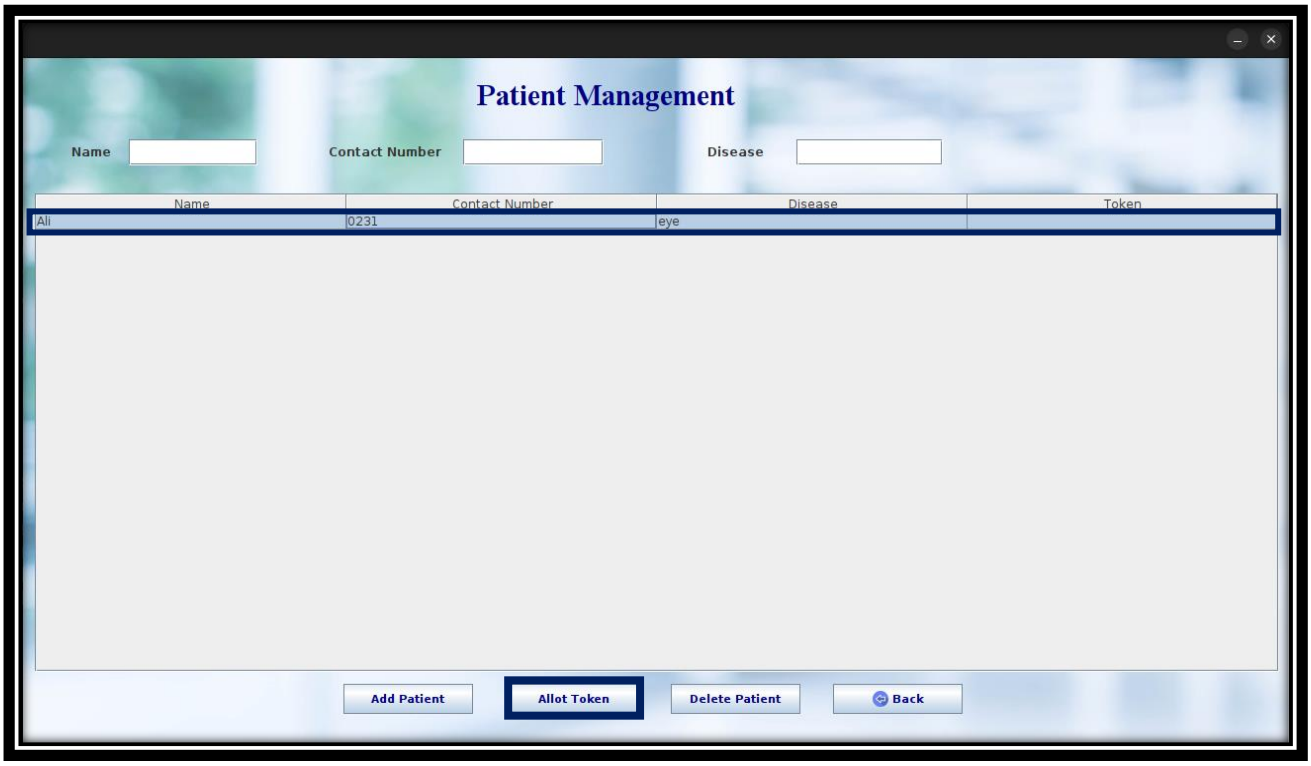
Adding information of patient



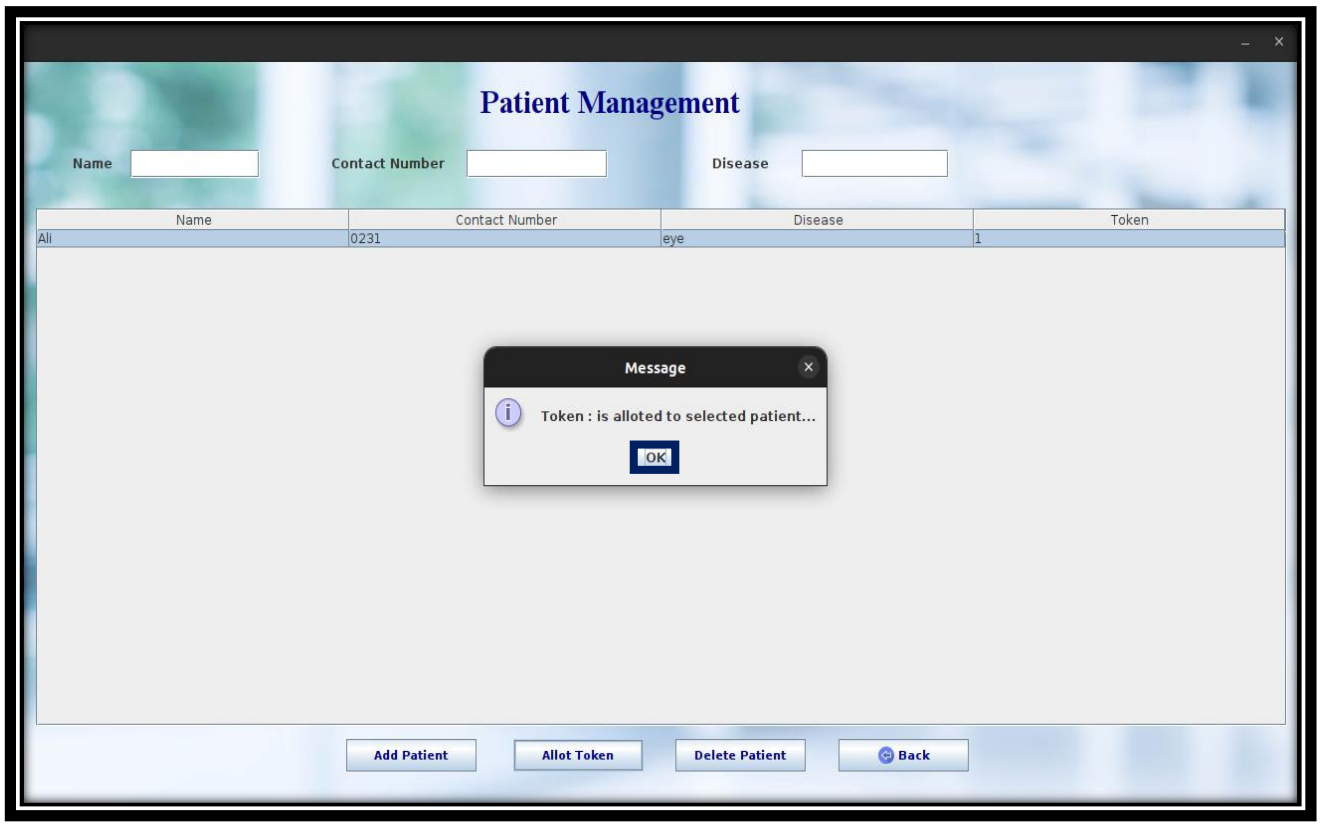
Add Patient Page



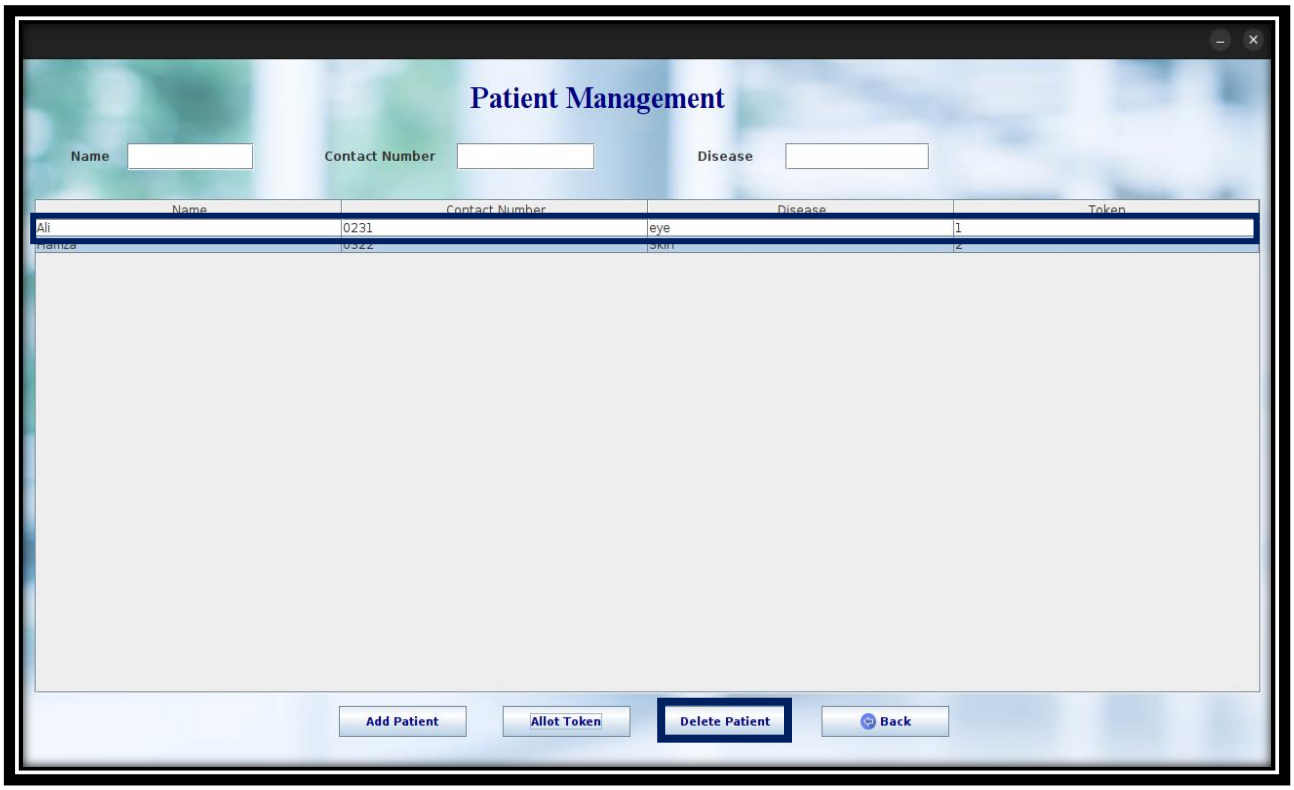
Selecting Patient to Allot token



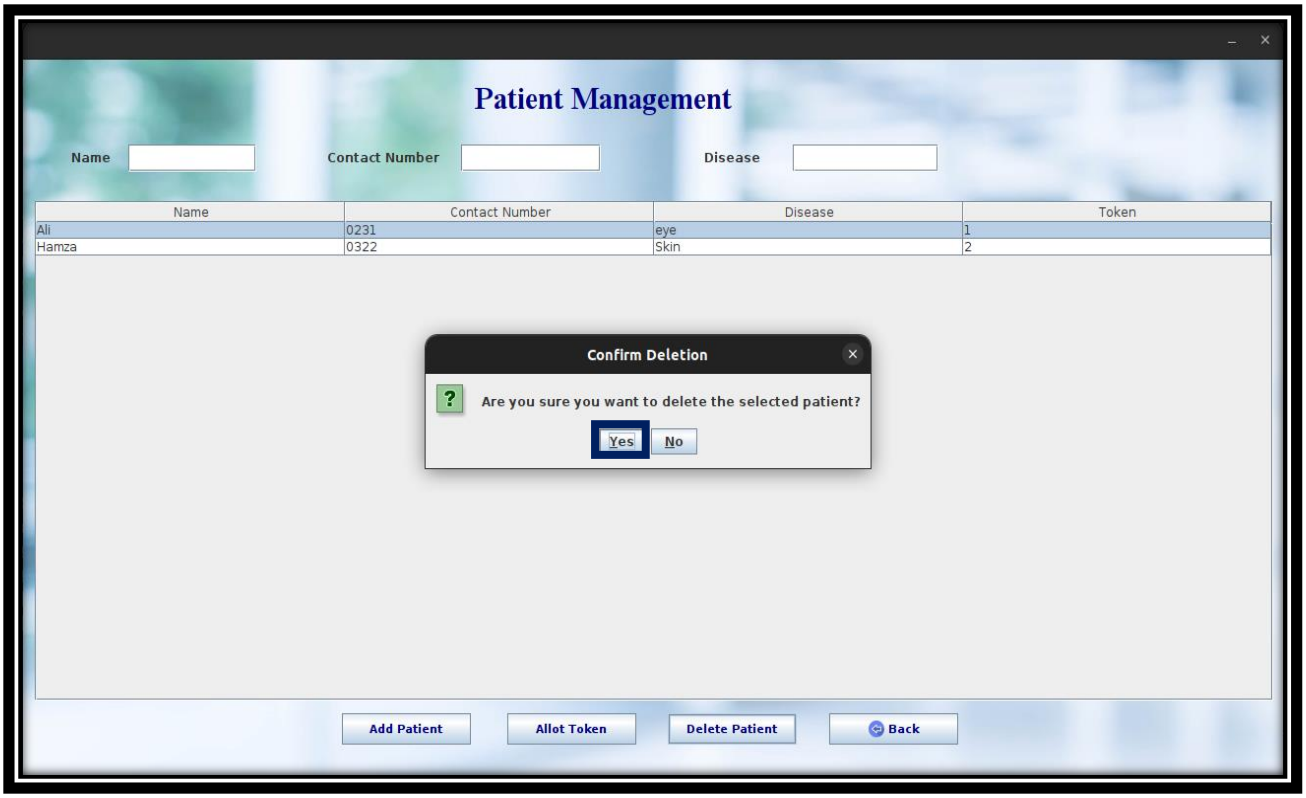
Allot Token Page



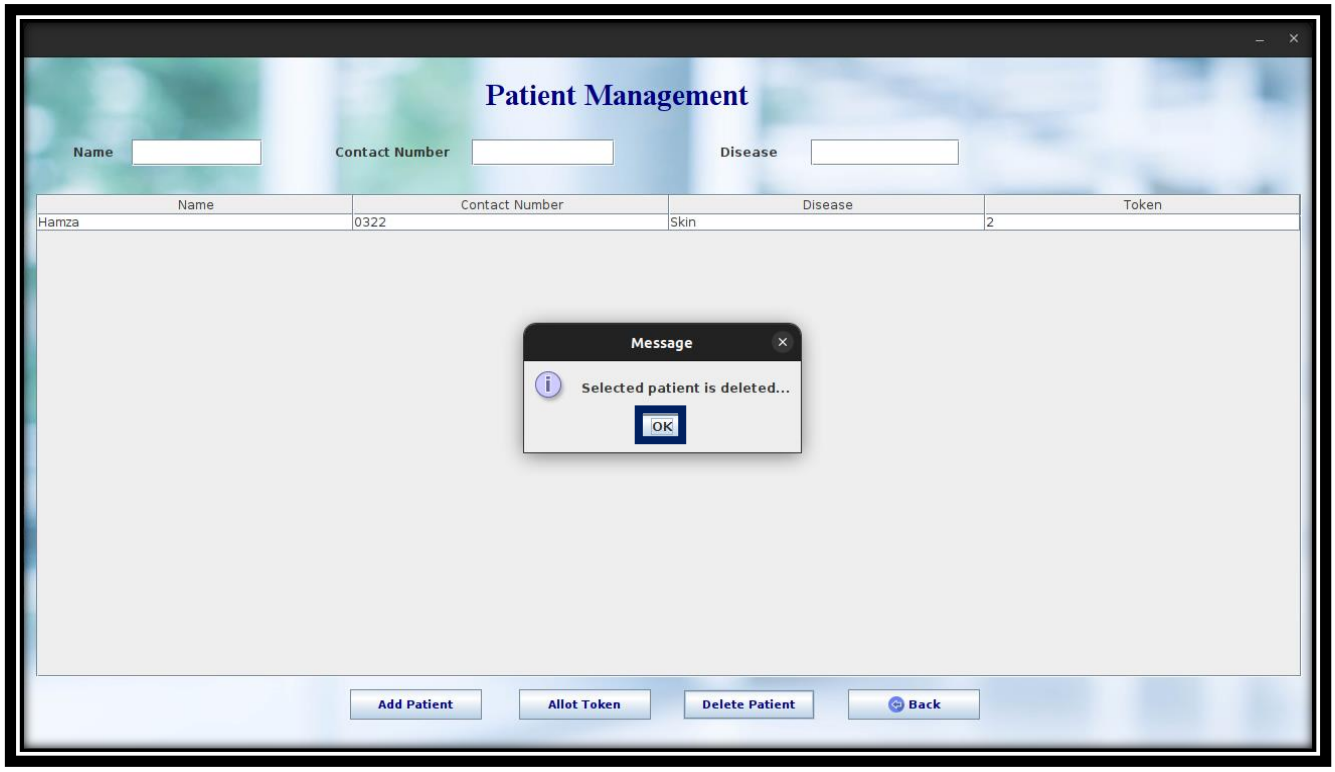
Selecting Patient to Delete



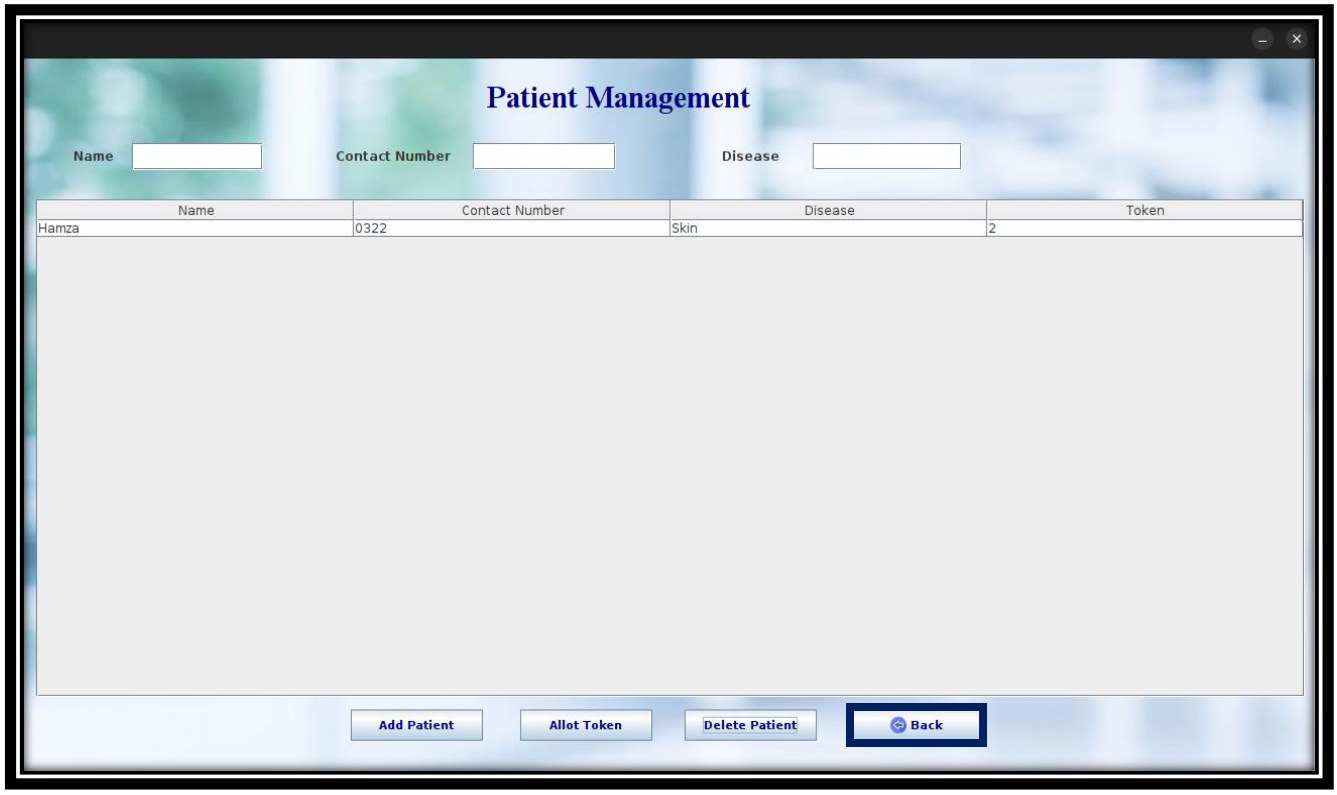
Confirmation Page



Delete Patient Page



Return to Main Page



Return to Home Page